

SE7596

MARIAN COLLEGE KUTTIKANAM (AUTONOMOUS)
UG Programme (Honors) Degree October 2024 Regular Examinations
First Semester

241CCRUBC101 Digital Fundamentals

Time: 2 Hours

Maximum Marks: 40

Answer one question from each bunch

BUNCH I

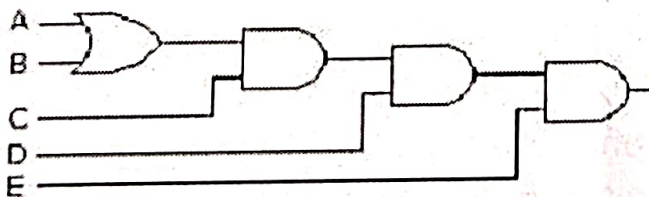
- I.A A digital watch sums times represented in BCD: 1001 0110 and 0011 0111. Perform the BCD addition and represent the result in BCD format. Explain the method used for addition. (CO1:8 Marks)

OR

- I.B A memory chip has a data size of 400 in hexadecimal. Convert this size into binary and decimal and explain the characteristics of hexadecimal representation. (CO1:8 Marks)

BUNCH II

- II.A.1 A smart home security system uses a circuit to determine whether to trigger an alarm. The circuit has five inputs: A and B represent signals from two independent motion sensors, C represents the status of a door sensor, D and E represent two separate window sensors. Derive the Boolean expression for the circuit.



(CO2:3 Marks)

- II.A.2 Emma is designing a control circuit for a smart lighting system in her home. The circuit has two inputs: **A**, representing whether the main light switch is on and **B**, representing whether the motion detector in the room detects movement. The output of the circuit is determined by the expression: $Y = A'B'(A'+B)(B'+B)$. Simplify the given Boolean expression using Boolean laws. (CO2:5 Marks)

OR

- II.B Anjali, a computer science student, is learning about logical circuit optimization and needs to understand the equivalence of expressions using DeMorgan's Theorems. To strengthen her understanding, she decides to prove the following rules using truth tables and logic diagrams:
1. $(X \cdot Y)' = X' + Y'$
 2. $(X + Y)' = X' \cdot Y'$
- II.B.1 Help Anjali by constructing truth tables and designing logic diagrams to prove these two rules, demonstrating how DeMorgan's Theorems can simplify and optimize logical circuits. (CO2:8 Marks)

BUNCH III

- III.A A digital security system uses a Boolean function $F = A + B'C$ to determine access permissions based on three input conditions: **A**, **B**, and **C**. How would you represent **F** as a sum of minterms to ensure all possible input combinations are accounted for in the logic circuit design? (CO3:8 Marks)

OR

- III.B Sophia is designing a cost-effective digital circuit and wants to implement the circuit using only **NAND gates** and **NOR gates**. Help her in the design process by showing how these universal gates can replace the basic gates **AND**, **OR** and **NOT**. (CO3:8 Marks)

BUNCH IV

- IV.A Liam is developing an irrigation control system where a single water flow signal needs to be directed to one of two sprinklers based on an external

control signal. The system requires a 1-to-2 demultiplexer to achieve this functionality.

- IV.A.1 Assist Liam by designing a 1-to-2 demultiplexer circuit using basic logic gates. Provide the truth table for the DEMUX and explain how the control signal determines which output line receives the input signal. Discuss how this design ensures efficient direction of water flow to the selected sprinkler. (CO4:8 Marks)

OR

- IV.B Sarah is developing a control system for a factory that has 16 different machines. Each machine can be turned on individually, but she wants to use a smaller digital control system to monitor which machine is active by encoding these signals into a compact 4-bit binary format. Design a 16-input encoder to track which machine is active and display the corresponding binary code? (CO4:8 Marks)

BUNCH V

- V.A.1 'Flip-flops are fundamental building blocks of any sequential circuit'. Is the argument correct? Justify your answer. (CO5:3 Marks)
- V.A.2 Meera is designing an automatic door controller that uses flip-flops to control open and close operations. The system must decide between edge triggering (door operates on clock transitions like rising or falling edges) and level triggering (door operates while the clock is HIGH or LOW). Differentiate between edge triggering and level triggering with the help of clock signals. (CO5:5 Marks)

OR

- V.B "JK flip-flop is a modified version of the SR flip-flop designed to avoid the undefined/invalid state of the SR flip-flop when both inputs are active."
- i. Is this statement correct? Justify your answer.
- ii. Explain the working of both SR flip-flop and JK flip-flop, and also provide the truth tables to support your explanation. (CO5:8 Marks)